PNAS RESEARCH ARTICLE | Social Sciences

Household structure, not clinical need, determines who absorbs the cost of long-term care

Alanna Polyak[a,1]

This manuscript was compiled on August 25, 2026

Long-term care in the United States is largely by unpaid family members, a transfer invisible to health expenditure accounting. Using the 2022 Health and Retirement Study, I constructed the share of monthly care hours a respondent receives from unpaid informal helpers ($n = 2{,}142$ recipients, $4{,}481$ helper records) and ask what predicts it. 84.7% of recipients receive all care from family or friends. Comparing ordinary least squares against cross-validated gradient boosting, a fractional-response specification, two-part models, and multilayer perceptrons, no model class explains more than a tenth of held-out variance; the neural network performs worse than predicting the sample mean. Across every specification, household composition dominates: living arrangement carries the largest coefficients and the largest SHAP attributions; stroke ($\beta = -0.015$, $P = 0.03$) and functional limitation ($\beta = -0.026$, $P = 0.001$) are statistically detectable but small. Interaction tests reveal care intensity and living alone interact ($\beta = -0.032$, $P = 0.005$), an effect recovered as the strongest pairwise SHAP interaction. A two-tier target-leakage in network features inflates held-out R^2 from 0.087 to 0.644. Automated US to English survey crosswalk achieved 71% top-1 accuracy versus a 0.2% random baseline. All analysis is descriptive; no causal effect is estimated.

informal caregiving | long-term care | machine learning | health equity | Health and Retirement Study

Significance

Most long-term care in the United States is unpaid work done by family members, and it appears in no measure of health spending. Using a nationally representative survey of 2,142 older adults who need help with daily activities, this study asks what determines whether that help comes entirely from relatives rather than from paid providers; the answer is not how sick someone is. Functional limitation and diagnosed neurological disease have small, barely detectable effects, while who a person lives with dominates every model tested. Read against the standard equity criterion for health services, which holds that need should drive utilization, this is a finding about inequity: the households least able to purchase an alternative are the ones absorbing the most unpaid labor.

Most long-term care for older Americans is unpurchased, provided by spouses and relatives often without training (1). Since this labor never transacts, it is absent from health expenditure accounts, rendering supplying households invisible to policy statistics. Among older adults who need help with basic daily activities, what determines whether that help arrives entirely from family, or whether paid and institutional providers enter?

Three frameworks make competing predictions. Andersen's behavioral model (2) partitions determinants into predisposing characteristics, enabling resources, and need; it supplies the standard equity criterion: in a well-functioning system need should dominate utilization, while reliance on enabling resources measures inequity. Cantor's hierarchical compensatory model (3) holds that older adults draw on a ranked ordering of caregivers(spouse, adult children, other kin, formal services), exhausting each tier sequentially; this predicts household composition dominates. Litwak's task-specific model (4) uniquely requires an interaction: informal and formal providers are matched to tasks by comparative advantage, meaning the effect of rising need depends on household availability.

0.0.0.1. Related work.. Cross-national comparisons establish that caregiving composition varies sharply with a country's long-term care system (5, 6), and microsimulation projects European informal care demand rising through 2050 (7). Work closest to this study examines how caregiving for adults with cognitive decline is distributed within and beyond the household (8). This paper extends that literature in three ways. First, treating the informal share as a continuous bounded outcome, rather than a binary indicator, shows how the resulting spike at one affects model choice. Second, it compares five model families under a design that respects household clustering, rather than reporting a single specification. Third, it treats the failure modes, leakage, a non-surviving hypothesis, and a failed automated harmonization, as results of their own.

Author affiliations: [a]Department of Quantitative Social Science, Dartmouth College, Hanover, NH 03755

A.P. designed the study, wrote all analysis code, performed the analysis, and wrote the paper.

The author declares no competing interests.

[1]To whom correspondence should be addressed. E-mail: alanna.g.polyak.28dartmouth.edu

Data

0.0.0.2. Source and window. The Health and Retirement Study (9) is a nationally representative biennial panel of US adults over 50 and their spouses, fielded by the Survey Research Center at the University of Michigan. This analysis uses the 2022 Core final release together with the 2022 Tracker file; the 2022 wave contains 15,787 respondent records collected by in-person, telephone and web interview, with proxy interviews taken where a respondent could not self-report. Proxy status matters because cognitive impairment is simultaneously a predictor of interest and a cause of proxy interviewing.

0.0.0.3. Unit of analysis. The original data are supplied at two levels. The respondent-level file records difficulty with six activities of daily living (ADLs: dressing, walking, bathing, eating, transferring, toileting) and five instrumental activities (IADLs: meal preparation, shopping, telephone use, medication management, money management). The *helper-level* file records one row per person who helps the respondent, carrying that helper's relationship, days of help per month, hours per helping day, whether insurance pays them, and out-of-pocket payment. The unit of analysis in this paper is the *respondent*; helper records are aggregated up.

0.0.0.4. Outcome construction. For each helper record, monthly care hours are days of help multiplied by hours per helping day, reconciling the three ways HRS permits the frequency question to be answered. A helper is classified as formal if the relationship is an organization, an institutional employee, a paid helper, a professional or an online service, and as informal otherwise: except that informal helpers whom the household reports paying out of pocket are reclassified as formal. That rule matters substantively: a daughter being paid to provide care is performing paid work, and counting her as informal would understate the market's role. Summing within respondent gives

$$s_i = \frac{\sum_{j \in \mathcal{I}_i} h_{ij}}{\sum_{j \in \mathcal{H}_i} h_{ij}}, \qquad [1]$$

where $\mathcal{H}_i$ is the set of helpers for respondent i, $\mathcal{I}_i \subseteq \mathcal{H}_i$ the informal subset, and h_{ij} monthly hours. After restricting to respondents aged 50 and over with usable hours data, the analysis sample is $n = 2{,}142$ with 33 base predictors.

0.0.0.5. Missingness. HRS encodes "don't know" and "refused" as reserved numeric values (8/9, 98/99, −8); these are converted to missing per variable rather than read as data. For every predictor with more than 1% missingness a binary indicator is created before median imputation; this is deliberate. Allowing a gradient-boosted model to route missing values down its own default branches would fold the fact of missingness into the SHAP attribution of the parent feature, which misleads when missingness is not at random, and here it is not. The word-recall test is missing for 21% of the sample, and missing among the cognitively impaired and proxy-interviewed respondents whose care arrangements are most distinctive.

0.0.0.6. Limitations of the source. Four are important to note. The design is cross-sectional, so nothing here speaks to how arrangements evolve as need progresses. Survey weights are available for only 80.3% of the sample, and critically for none of the 73 nursing-home residents, because HRS respondent weights are constructed for the community-dwelling population; any weighted analysis therefore silently drops institutionalized respondents. Section G asks about helpers who assist the respondent personally, and facility staff appear under-reported relative to family visits, which produces a counterintuitive positive association between nursing-home residence and the informal share. Finally, the outcome is constructed from the helper roster, so any feature derived from that roster is downstream of the outcome: a point returned to below.

Methods

0.0.0.7. Cleaning. `code/00_build_analysis_file.py` performs every cleaning step. HRS reserved codes for "don't know" and "refused" are mapped to missing before any calculations, via `clean_codes` in `utils.py`; leaving them in place would enter values such as 998 into an hours total. Helper records with no usable hours or no relationship code are dropped (4,481 records to 3,666). Helpers paid out of pocket by the family are reclassified from informal to formal, because the analytic distinction is whether the household bears the cost, not whether the helper is a relative.

0.0.0.8. Merging. Three joins, all exact on the composite key (HHID, PN), which uniquely identifies a respondent in HRS; no unclear matching is used anywhere. The helper file is aggregated to respondent level and **inner joined** to the respondent file, then **left joined** to the tracker file for survey weights and household identifiers; the script prints row counts before and after each join. Table 1 reports the resulting attrition.

0.0.0.9. Variable definitions. The outcome is

$$s_i \;=\; \frac{\sum_{h \in \mathcal{I}_i} \mathrm{hours}_{ih}}{\sum_{h \in \mathcal{H}_i} \mathrm{hours}_{ih}}, \qquad [2]$$

where $\mathcal{H}_i$ is respondent i's full helper roster and $\mathcal{I}_i \subseteq \mathcal{H}_i$ the unpaid informal helpers, so $s_i \in [0, 1]$. Predictors fall into four blocks: demographics (age, sex, years of schooling, race); household structure (lives alone, with partner, with relative; number of children); clinical need (ADL count, IADL count, self-rated health, stroke, dementia, word recall, memory rating); and care intensity (total monthly hours, number of helpers).

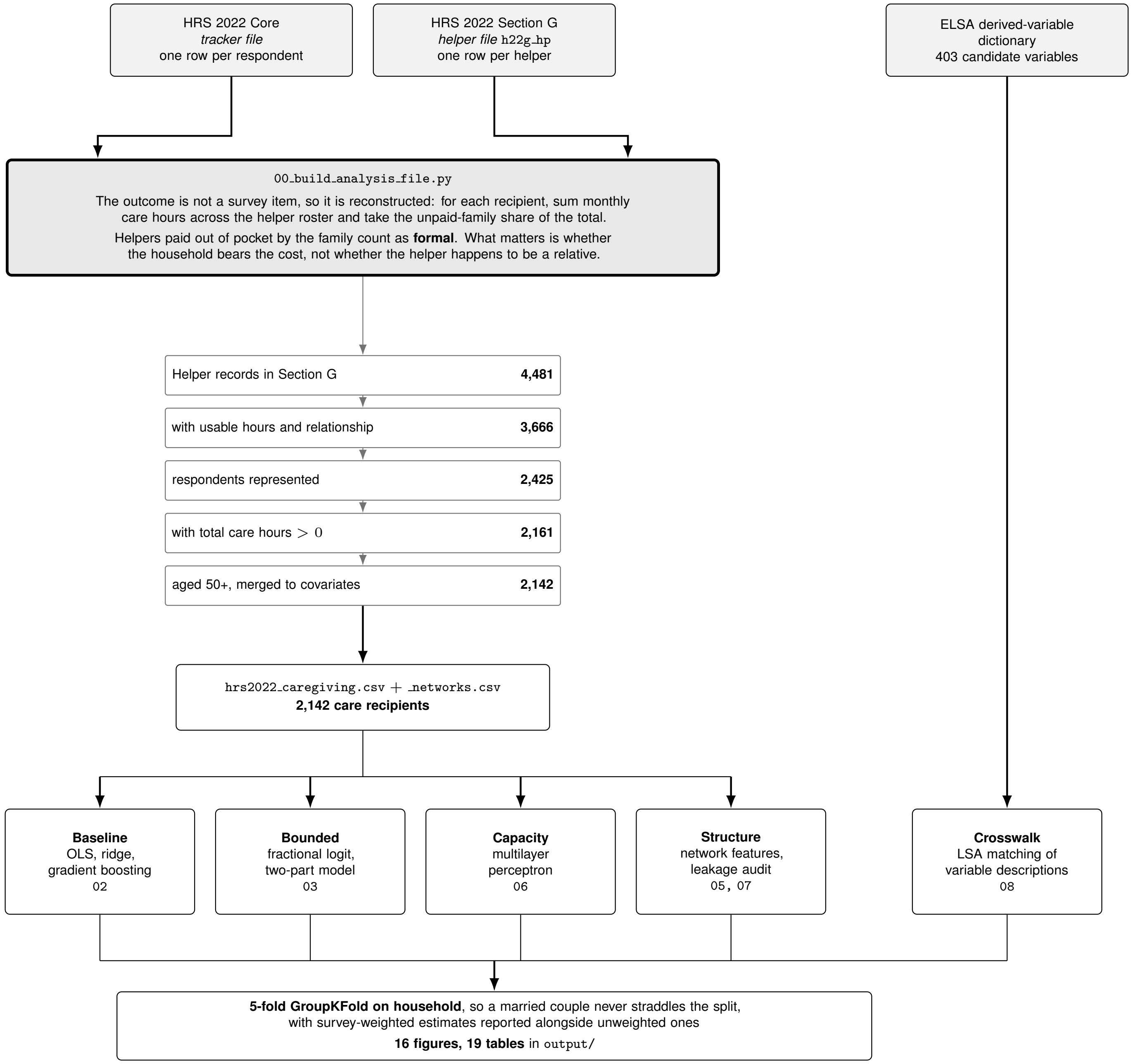


Fig. 1. The analysis pipeline, in the order it runs. Grey boxes are sources taken as given; the large box is where the outcome is constructed, and that construction is what makes the leakage audit in 07 necessary: every feature derived from the helper roster is downstream of the outcome by design, not by accident. The ladder reports the sample remaining at each filtering step.

0.0.0.10. Train-test split and validation. Two schemes are reported because they disagree. A single 25% held-out split (`random_state` = 20260818) is used in `code/02_baseline_ols_vs_gbm.py` for the model-family comparison, with booster hyperparameters selected by an inner five-fold grid search on the training portion only, so the test split is evaluated once. All other performance figures use five-fold cross-validation grouped on household, which is the design adopted from `code/04_weights_and_clustered_cv.py` onward. Every random seed is fixed at 20260818.

0.0.0.11. Identification. This is a descriptive and predictive analysis. There is no randomization, instrument or discontinuity, and living arrangement is not randomly assigned; it is as plausibly a consequence of care need as a determinant of how that need is met. No causal effect is estimated or implied anywhere in this paper.

0.0.0.12. Design. All held-out performance is estimated by five-fold cross-validation grouped on household, so that no household spans the train/test boundary. Spouses are both interviewed in HRS, and 5.3% of households contribute more than one respondent, covering about 10% of rows. Standard errors on interaction coefficients are clustered on household to match.

0.0.0.13. Model families. Five specifications are compared on identical folds and identical features: (i) ordinary least squares on standardized predictors; (ii) a gradient-boosted regressor (XGBoost) (10) with depth, tree count, shrinkage and minimum child weight selected by nested five-fold cross-validation on the training split; (iii) a fractional-response

GLM (11) with a binomial family and logit link, which uses the binomial likelihood as a quasi-likelihood and guarantees predictions inside $[0, 1]$; (iv) an explicit two-part model, a logit for whether any formal care is present followed by a conditional model for the share among those who use some, with combined prediction

$$\mathbb{E}[s_i \mid x_i] = \big(1 - \pi(x_i)\big) \cdot 1 + \pi(x_i)\, \mathbb{E}[s_i \mid s_i < 1, x_i], \qquad \textbf{[3]}$$

where $\pi(x_i) = \Pr(s_i < 1 \mid x_i)$; and (v) a multilayer perceptron with architecture and L_2 penalty selected by grid search.

The fractional-response specification models the conditional mean directly through a logit link,

$$\mathbb{E}[s_i \mid x_i] \;=\; \Lambda(x_i^\top \beta) \;=\; \frac{\exp(x_i^\top \beta)}{1 + \exp(x_i^\top \beta)}, \qquad \textbf{[4]}$$

estimated by maximizing the Bernoulli quasi-likelihood $\sum_i \{s_i \log \Lambda(x_i^\top \beta) + (1 - s_i) \log[1 - \Lambda(x_i^\top \beta)]\}$. The estimator stays consistent for β even though s_i is a proportion rather than a binary outcome, and predictions are confined to $(0, 1)$ by construction.

0.0.0.14. Interpretation. Linear models are interpreted through standardized coefficients with 95% confidence intervals; tree models through SHAP values (12), which decompose each individual prediction into per-feature contributions. These are different quantities: a coefficient is a partial effect identical for every respondent, a SHAP value is a marginal contribution that varies across them; the paper treats their disagreement as informative. Pairwise SHAP interaction values are used to detect interactions the model used, alongside pre-specified product terms.

Results

0.0.0.15. The analysis sample. Table 1 characterizes it and records where each step of attrition occurs.

Table 1. Construction of the analysis sample, HRS 2022 Core. Each row is the count surviving that step. The largest single loss is helper records missing usable hours or a relationship code, which cannot contribute to the outcome in Eq. 2. Respondents receiving no measurable care hours are dropped because the outcome is undefined at a zero denominator, not because they are uninformative.

Step	n
Helper records in HRS 2022 Section G	4,481
with usable hours and relationship	3,666
Respondents represented	2,425
with total care hours > 0	2,161
aged 50+, merged to covariates	**2,142**

Single survey wave, fielded 2022. Analysis sample 2,142 respondents; 15.4% are proxy interviews and 73 are nursing-home residents who carry no survey weight.

The outcome is nearly degenerate. Figure 2 establishes the constraint that governs everything after it. Of respondents receiving any help, 84.7% receive every hour of it from unpaid family and friends; 5.9% receive none informally; the intermediate range is nearly empty. Median care intensity is 52 hours per month with a long right tail, and respondents have on average 1.7 helpers. Care arrangements in this population are close to binary: either the family absorbs the entire load or formal providers are involved; there is correspondingly little variance for any model to explain.

Daughters are the modal caregiver (1,100 helper records), ahead of spouses and partners (924) and sons (577); paid helpers account for 243. Daughters outnumber sons roughly two to one, a gendered pattern the hierarchical compensatory model does not itself predict.

Household structure dominates clinical need. The mean informal share declines from 0.928 among respondents with no ADL limitations to 0.837 among those with four, then flattens and partially reverses: the most severely limited respondents are not the most likely to have formal help. Respondents with a stroke, dementia or Alzheimer's diagnosis have a mean informal share of 0.888 against 0.900 for those without: a difference of roughly one percentage point.

Against this, living arrangement carries the largest effects in every specification. In the linear model, living with a partner ($\beta = +0.061$, $P = 0.006$) and living with a relative ($\beta = +0.046$, $P = 0.012$) are the largest coefficients, while ADL count ($\beta = -0.026$, $P = 0.001$) and stroke ($\beta = -0.015$, $P = 0.032$) are detectable but an order of magnitude smaller. Read through Andersen's framework this is an equity result rather than a null one: enabling structure, not need, is allocating care.

Coefficients and SHAP disagree, for identifiable reasons. The Spearman correlation between the OLS importance ranking and the SHAP ranking is $\rho = 0.157$ (Fig. 3); two features drive the disagreement. Living alone ranks first by SHAP and 24th by coefficient magnitude, because the three living-arrangement indicators nearly partition the sample: in the linear model they compete for a shared signal against an omitted reference category and the coefficient mass distributes arbitrarily among them, while the tree splits on whichever single indicator separates best and credits that one. Total care hours ranks second by SHAP and 31st of 33 by coefficient, because its relationship to the outcome is non-monotone: very low-intensity help is almost always informal, very high-intensity help skews formal, the middle is mixed; thus, a single linear slope averages to nearly nothing while the tree captures it with a few splits.

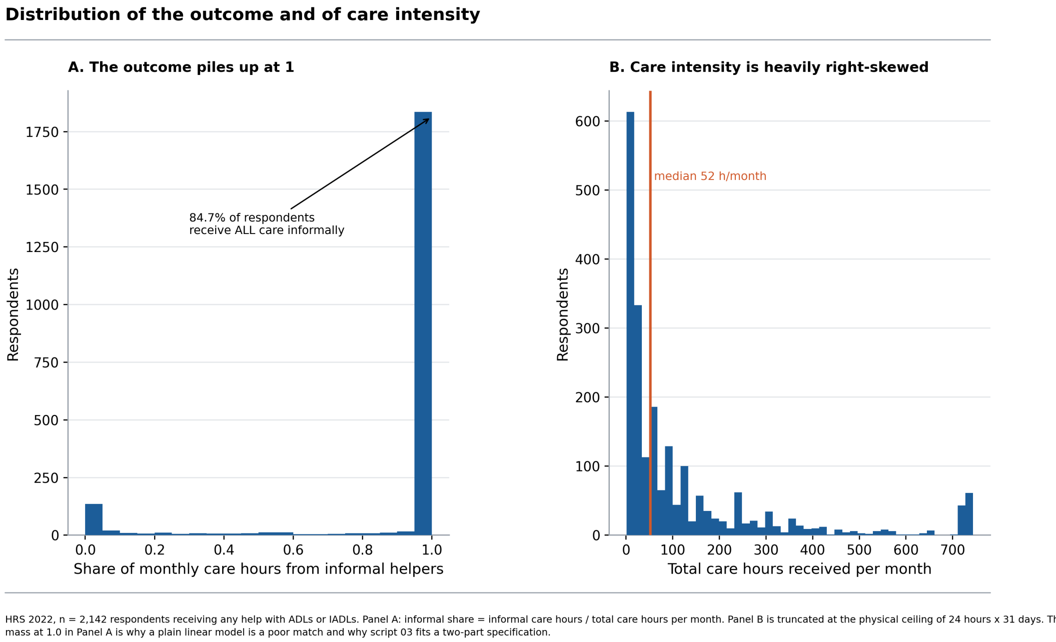


Fig. 2. Distribution of the informal care share and of care intensity. HRS 2022, $n = 2{,}142$ respondents receiving any help with ADLs or IADLs.

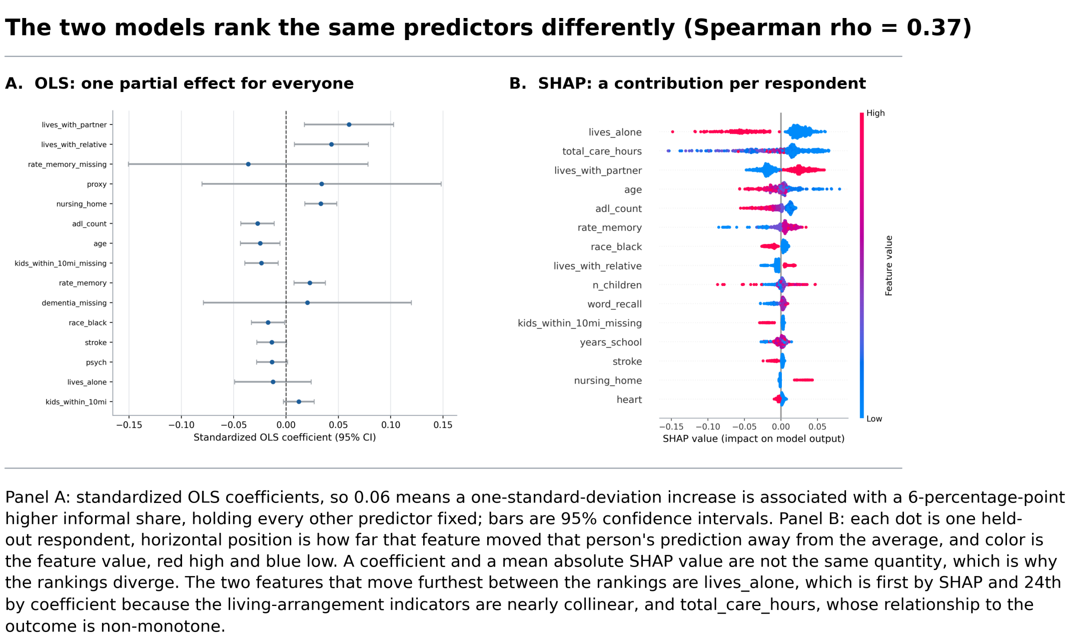


Fig. 3. Standardized OLS coefficients with 95% confidence intervals against the SHAP beeswarm for the tuned gradient-boosted regressor. The two methods rank the same predictors differently ($\rho = 0.37$).

No model class does well, and the neural network does worst. Table 2 reports both evaluation schemes, because they disagree. On a single 25% split gradient boosting reaches $R^2 = 0.039$ against 0.011 for OLS; under household-grouped five-fold cross-validation the booster reaches 0.087 and a ridge model 0.061. A single split on 2,142 rows with an 85% point mass is optimistic about the gap between model classes and pessimistic about their level. Tuning does the work: an untuned booster scores -0.053, worse than predicting the mean, so reporting only the tuned figure would misrepresent boosting as reliably superior at this sample size.

The multilayer perceptron scores below the intercept-only baseline. An architecture search over seven configurations from a single eight-unit layer to three stacked layers finds no configuration that beats the mean, and the best is the smallest; an L_2 sweep over five penalties points the same: cross-validated performance improves monotonically as the penalty rises toward the linear limit, from -0.17 at $\alpha = 10^{-4}$ to -0.001 at $\alpha = 1$. Both are the signature of a model with far more parameters than 2,142 rows can identify.

The result was predicted before fitting, which is what makes it informative. Boosted trees are the default on small tabular data; neural networks are helpful for high-dimensional structured inputs, none of which is present here.

Table 2. **Held-out performance; regression scored by R^2, classifiers by AUC. The lower panel uses five-fold cross-validation grouped on household, so a married couple never straddles the train and test sets. No model class explains more than about a tenth of the variance in the informal care share.**

Model	Task	Held-out
Single 25% held-out split		
Intercept only	informal share	-0.007
OLS	informal share	0.011
XGBoost, untuned defaults	informal share	-0.053
XGBoost, CV-tuned	informal share	$\mathbf{0.039}$
Fractional logit	informal share	0.032
Two-part model	informal share	$\mathbf{0.035}$
Logistic regression	any formal care	0.710
XGBoost classifier	any formal care	$\mathbf{0.728}$
Five-fold CV, grouped on household		
XGBoost, CV-tuned	informal share	$\mathbf{0.087}$
Ridge (linear)	informal share	0.061
Intercept only	informal share	-0.000
Multilayer perceptron	informal share	-0.001

A bounded outcome needs a bounded likelihood. OLS produces 58 held-out predictions outside $[0, 1]$, which are not interpretable as shares. Both the fractional logit and the two-part model of Eq. 3 are constrained by construction and produce none, at comparable predictive accuracy (Fig. 4). Decomposing the two-part model is informative in itself: the first stage, predicting whether any formal care is present, reaches AUC 0.710 and is well calibrated, while the second stage, how much of the load formal care takes once present, is close to unpredictable from these covariates. The covariates tell you whether the market enters, not how far it goes.

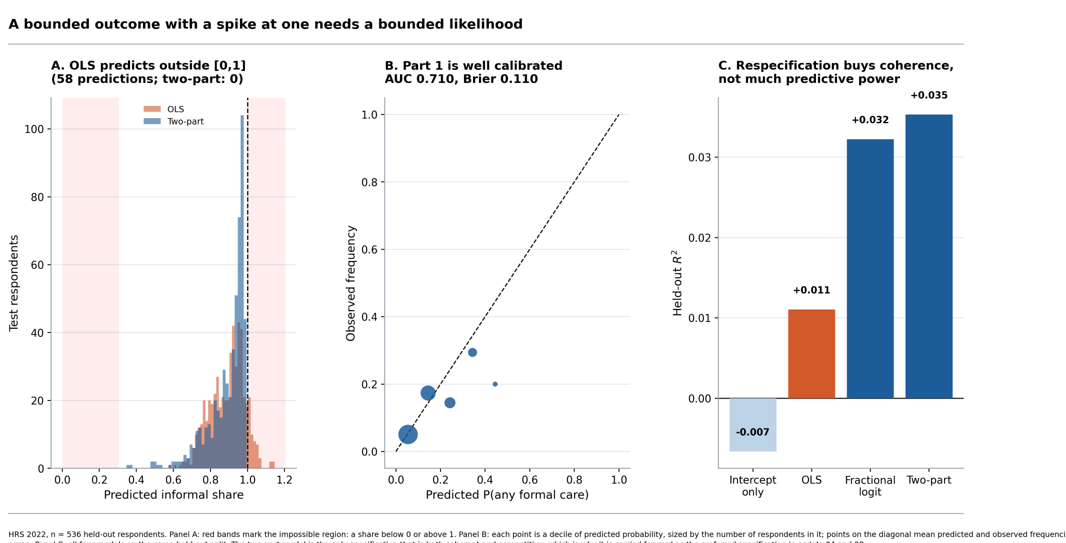


Fig. 4. A bounded outcome with a spike at one requires a bounded likelihood. Red bands in the left panel mark the impossible region.

Need and living arrangement interact. Litwak's model is the only one of the three that requires an interaction, and main-effects models assume it away before they start. Fourteen pre-specified interaction tests were run, of which five are significant at the 5% level against roughly 0.7 expected by chance. The strongest is between total care hours and living alone ($\beta = -0.032$, $P = 0.005$): among people living alone, rising care intensity pushes arrangements toward formal provision far more sharply than it does for people living with a partner. Independently, the pairwise SHAP interaction decomposition ranks the same pair first out of 630 (Fig. 5). Two methods with different assumptions selecting the same interaction is stronger evidence than either alone.

The within-group need gradients are also revealing. The slope of the informal share on ADL count is -0.011 ($P = 0.0006$) among respondents living with a partner and -0.014 ($P = 0.031$) among those living with a relative, but a flat and insignificant -0.005 ($P = 0.52$) among those living alone; for people who live alone, additional functional limitation does not systematically bring formal care in: which is the opposite of what an equity-satisfying system would produce.

Table 3 lists them. The pattern is coherent rather than scattered: care intensity and instrumental-activity need both interact with living arrangement, and the two living-arrangement moderators carry opposite signs, as they must if they are picking up the same underlying contrast from opposite ends. With fourteen tests roughly 0.7 significant

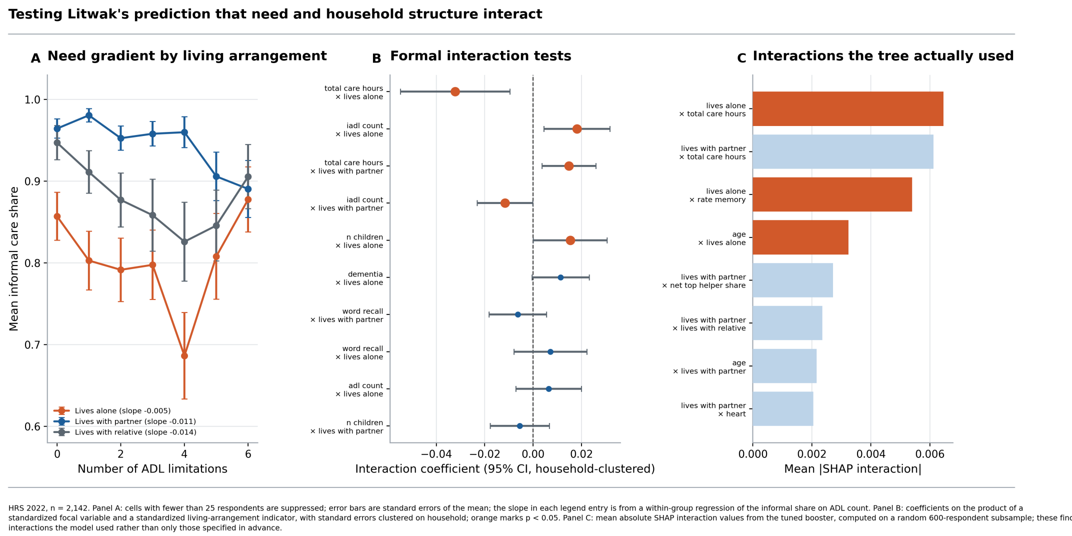


Fig. 5. Testing the prediction that need and household structure interact. Panel B shows pre-specified interaction coefficients with household-clustered standard errors; panel C shows interactions the tree used.

Table 3. The five interactions significant at the 5% level, of fourteen pre-specified tests. Standard errors are clustered on household.

Focal variable	Moderator	β (P)
Total care hours	Lives alone	-0.032 (0.005)
IADL count	Lives alone	$+0.018$ (0.009)
Total care hours	Lives with partner	$+0.015$ (0.009)
IADL count	Lives with partner	-0.012 (0.048)
Number of children	Lives alone	$+0.015$ (0.048)

results would be expected by chance, so five is well above the multiple-testing floor, though the individual P-values should not be read as if each test stood alone.

A two-tier leakage failure. Treating each respondent's helper set as an ego network yields nine structural features. Two are the outcome re-encoded: `net_any_formal` is true exactly when the informal share is below one, and including them lifts cross-validated R^2 from 0.087 to 0.644; removing them is not sufficient. Four more being removed, the kin-composition features, are zero when nobody in the roster is family, which is the same event as the informal share being zero: 94 respondents have zero distinct kin types and their mean informal share is 0.000. That second tier still yields $R^2 = 0.553$, and is invisible to a correlation scan because each feature looks unremarkable in isolation. With both tiers removed, the exogenous network features move R^2 from 0.0874 to 0.0864, which is essentially not at all. Figure 13 shows both tiers: the ladder of R^2 across the three feature sets, and the mechanism that makes the first tier a re-encoding of the outcome rather than a predictor of it.

The general point is that the outcome is constructed from the helper roster, so every roster-derived feature is downstream of it. The operative test is not whether a feature correlates with the outcome but whether it could have been known without knowing the outcome.

Automated cross-national harmonization. Extending this design across countries requires mapping HRS variables onto their counterparts in sister surveys, and that mapping is where there may be a stall: the English Longitudinal Study of Ageing calls the dressing item `headldr` where HRS calls it `SG014`. Embedding the variable descriptions and matching on cosine similarity reconstructs the crosswalk at 71.4% top-1 and 78.6% top-5 accuracy against all 403 ELSA derived variables, where random selection would score 0.2% (Fig. 16).

The failure pattern is the useful part; on ten ADL and IADL items the matcher gets nine right. On four items where the two studies measure the same construct with different instruments it gets zero right; e.g. matching "number of years in school" to ELSA's count of adults in the benefit unit at 0.82 similarity. However, all four errors fall in the bottom four by similarity margin, the gap between the best and second-best candidate; this is a rule that auto-accepts high-margin matches and routes the rest to a human that would have caught the errors while accepting 71% of the crosswalk automatically. A matcher that is wrong 29% of the time is not deployable; one that is wrong 29% of the time and knows which 29% is.

Discussion

Across five model families, three outcome specifications, survey weighting, household-clustered validation, network features and an interaction analysis, the substantive conclusion does not change: household composition determines who absorbs the cost of long-term care, and clinical need barely registers; that stability is the most useful thing this quantity of robustness can report.

The equity reading is direct: Andersen's criterion holds that need should drive utilization and that enabling resources driving it instead is the signature of inequity; here enabling structure dominates, and the interaction results sharpen the point: among people living alone, who by construction have no household member to absorb care, additional functional limitation does not reliably bring formal provision in. The households with the least internal capacity to absorb care are not being compensated by the market.

Four limitations bound the study; moreover, nothing here is a causal estimate. The analysis is cross-sectional and single-country; the outcome's degeneracy caps achievable accuracy for every model class, so absolute performance is not a verdict on the methods; living arrangement is not randomly assigned and is plausibly a consequence of care need as much as a determinant of how it is met; and the nursing-home weighting artifact means institutionalized respondents belong in a separate model.

Materials and Methods

All code is available at https://github.com/alannap27/qss45_informal_care/tree/main as ten numbered Python scripts with shared utilities in `code/utils.py` and figure layout rules in `code/figstyle.py`. HRS microdata are not redistributable and require free registration at https://hrs.isr.umich.edu; the repository documents the rebuild path. Analyses used Python 3.10 with pandas, scikit-learn, statsmodels, XGBoost, SHAP and NetworkX; exact versions are pinned in `requirements.txt`. Random seeds are fixed throughout.

Data, Materials, and Software Availability. Every figure and table in this paper, and the twelve figures and sixteen tables not shown, are regenerated by `bash run_all.sh` once the HRS files are in place. Script 00, which rebuilds the analysis file from the raw Section G helper records, is excluded from that command because its input cannot be redistributed.

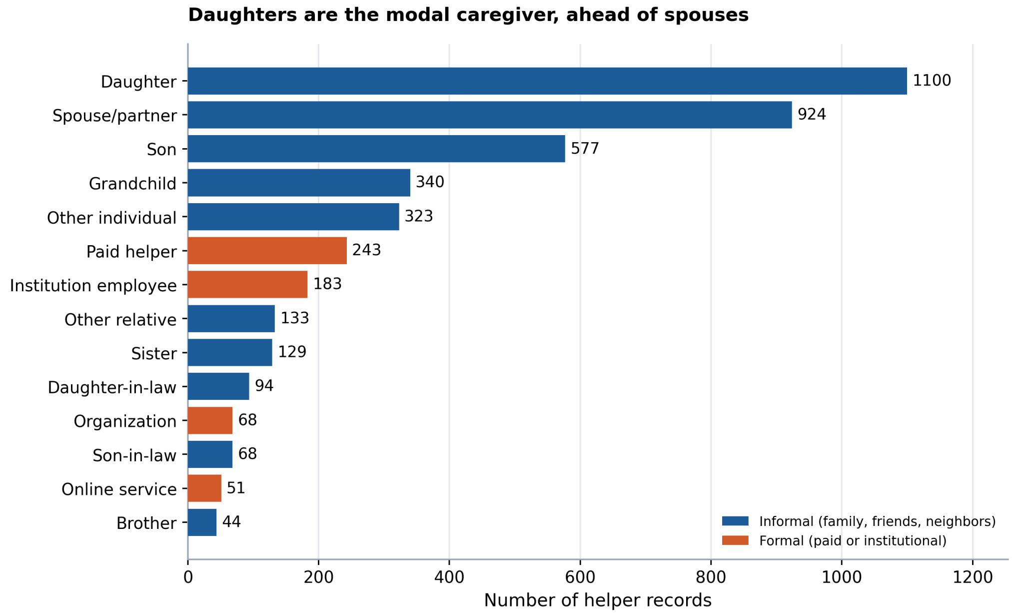
ACKNOWLEDGMENTS. I thank Prof. Herbert Chang for guidance on the analysis design, and the QSS 45 cohort for feedback on an earlier version of these results.

1. Alzheimer's Disease International, Karolinska Institutet, Global estimates of informal care, (Alzheimer's Disease International), Technical report (2018).
2. RM Andersen, Revisiting the behavioral model and access to medical care: Does it matter? *J. Heal. Soc. Behav.* **36**, 1–10 (1995).
3. MH Cantor, Neighbors and friends: An overlooked resource in the informal support system. *Res. on Aging* **1**, 434–463 (1979).
4. E Litwak, *Helping the Elderly: The Complementary Roles of Informal Networks and Formal Systems.* (Guilford Press, New York), (1985).
5. SN Jang, M Avendano, I Kawachi, Informal caregiving patterns in korea and european countries: a cross-national comparison. *Asian nursing research* **6**, 19–26 (2012).
6. OECD, Health at a glance 2023: Oecd indicators, (OECD Publishing, Paris), Technical report (2023).
7. A Cattaneo, A Vitali, D Regazzoni, C Rizzi, The burden of informal family caregiving in europe, 2000–2050: a microsimulation modeling study. *The Lancet Reg. Heal. – Eur.* **53**, 101295 (2025).
8. SS Matina, et al., Distribution of informal caregiving for older adults living with or at risk of cognitive decline within and beyond family in rural south africa. *The Journals Gerontol. Ser. B* **80**, gbaf008 (2025).
9. Health and Retirement Study, Health and retirement study, 2022 core final release (2024) Public use dataset. Produced and distributed by the University of Michigan with funding from the National Institute on Aging (grant number NIA U01AG009740). Ann Arbor, MI.
10. T Chen, C Guestrin, XGBoost: A scalable tree boosting system in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.* pp. 785–794 (2016).
11. LE Papke, JM Wooldridge, Econometric methods for fractional response variables with an application to 401(k) plan participation rates. *J. Appl. Econom.* **11**, 619–632 (1996).
12. SM Lundberg, SI Lee, A unified approach to interpreting model predictions in *Advances in Neural Information Processing Systems 30.* pp. 4765–4774 (2017).

Appendix A. The remaining figures

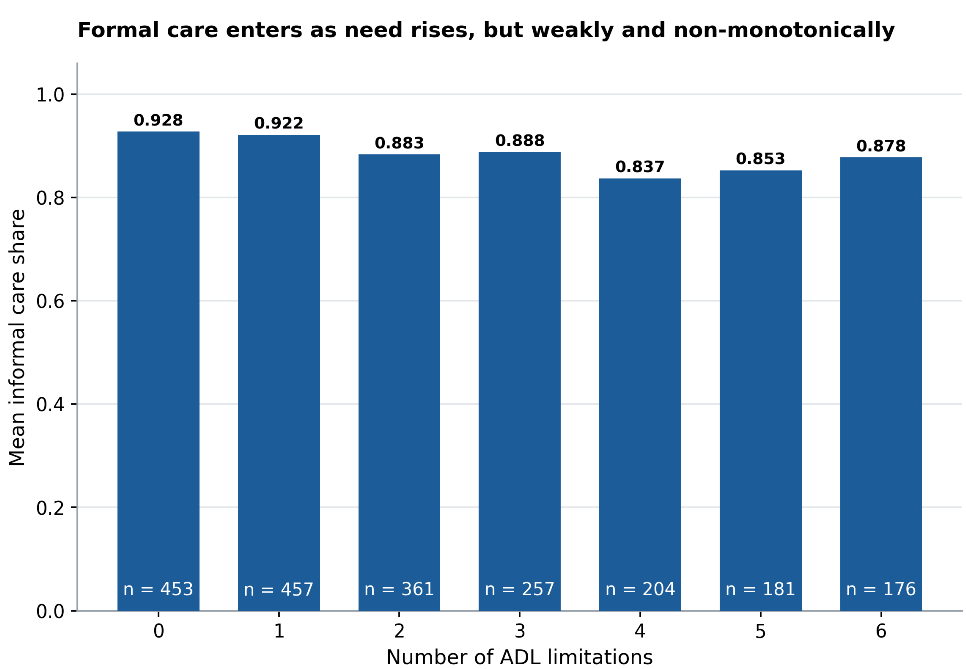
Four figures and three tables appear in the body. The pipeline produces sixteen figures and nineteen tables; the twelve held back are reproduced here with the captions they carry in the repository.

A.1 Description.



All 4,481 helper records in HRS 2022 Section G, before the out-of-pocket-payment reclassification applied in script 00. Daughters (1,100) outnumber sons (577) roughly two to one, a gendered pattern the hierarchical compensatory model does not itself predict. Formal providers of all kinds together account for a small minority of records.

Fig. 6. Who the helpers are, across all 4,481 helper records in HRS 2022 Section G. Daughters outnumber sons roughly two to one, which is the pattern the caregiving literature would predict and is the closest thing here to a validation of the reconstructed roster.



Mean informal care share by count of the six HRS activities of daily living (dressing, walking, bathing, eating, transferring, toileting). Cells with fewer than 30 respondents are suppressed. The share falls from 0.928 at zero limitations to 0.837 at four, then partially reverses: the most severely limited respondents are not the most likely to have formal help.

Fig. 7. The unconditional need gradient, before it is split by living arrangement. Read alone this looks like need barely matters; only splitting it, as in Fig. 5, reveals that need matters a great deal for some households and not at all for others. The two figures are the same data.

A.2 Model comparison.

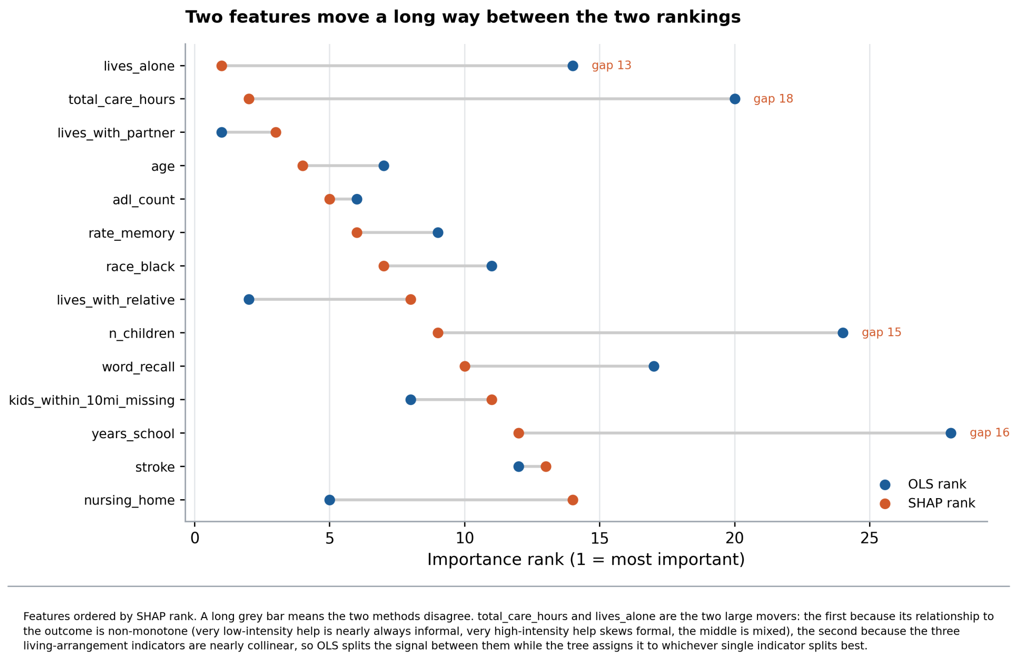


Fig. 8. Importance-rank agreement across four methods. The coefficient-against-SHAP contrast in the body is one slice of this. Methods that answer different questions rank predictors differently, and the spread here is the reason a single importance ranking should not be reported on its own.

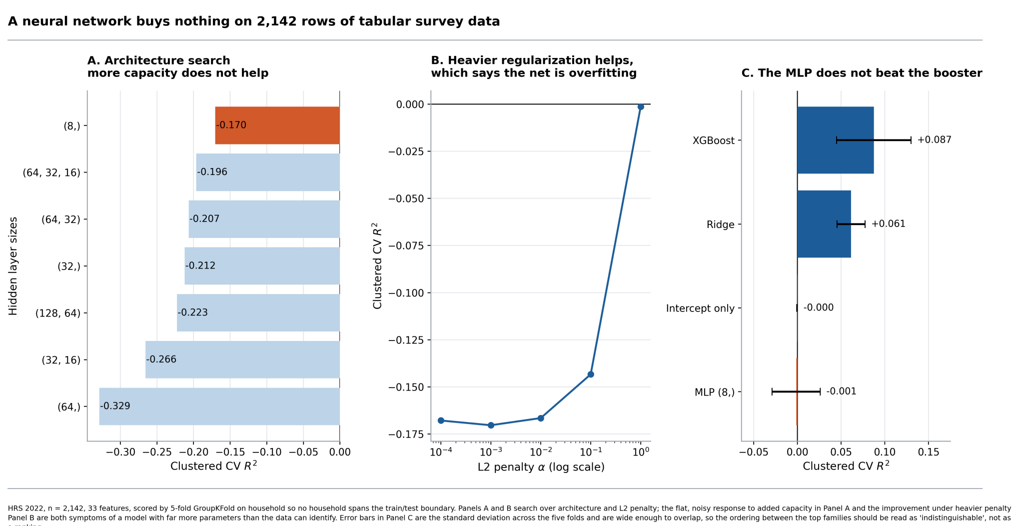


Fig. 9. A neural network buys nothing on 2,142 rows of tabular survey data. Panel A searches seven architectures and finds none that beats the mean, with the best being the smallest. Panel B sweeps five L_2 penalties and improves monotonically toward the linear limit. Both are the signature of a model with far more parameters than the data can identify.

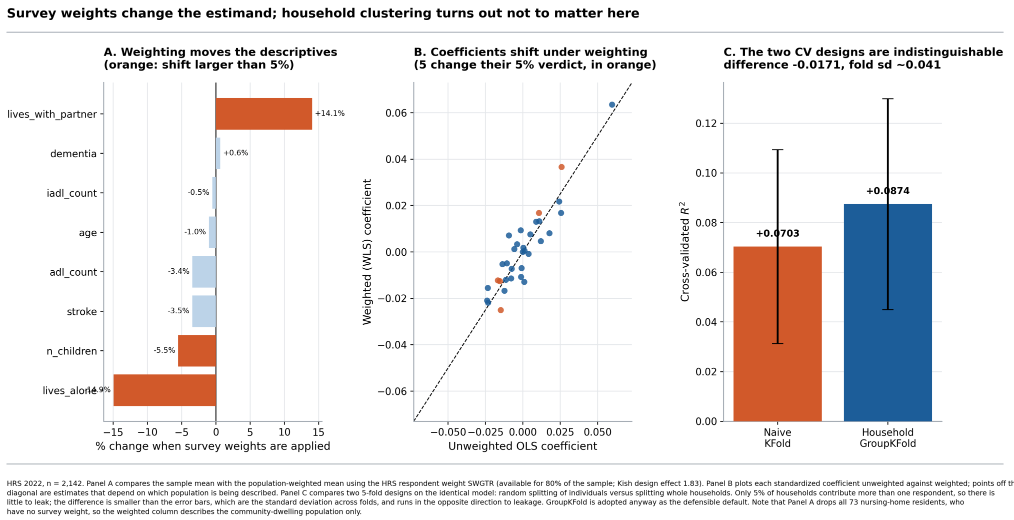


Fig. 10. Survey weights and household clustering. Weighting changes the estimand, because all 73 nursing-home residents carry no weight and are silently dropped. Household-grouped cross-validation scores higher than naive splitting, which is the opposite of the expected direction and is reported as such.

A.3 Care networks.

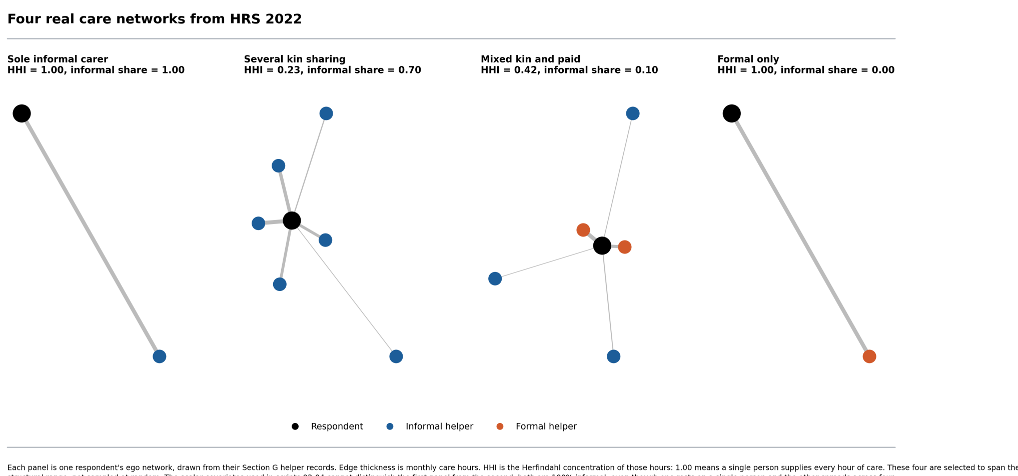


Fig. 11. Four real care networks, drawn from Section G helper records and selected to span the structural range rather than sampled at random. Edge thickness is monthly care hours. The scalar covariates used in scripts 02–04 cannot distinguish the first panel from the second: both are 100% informal, but one rests on a single person and the other spreads across four.

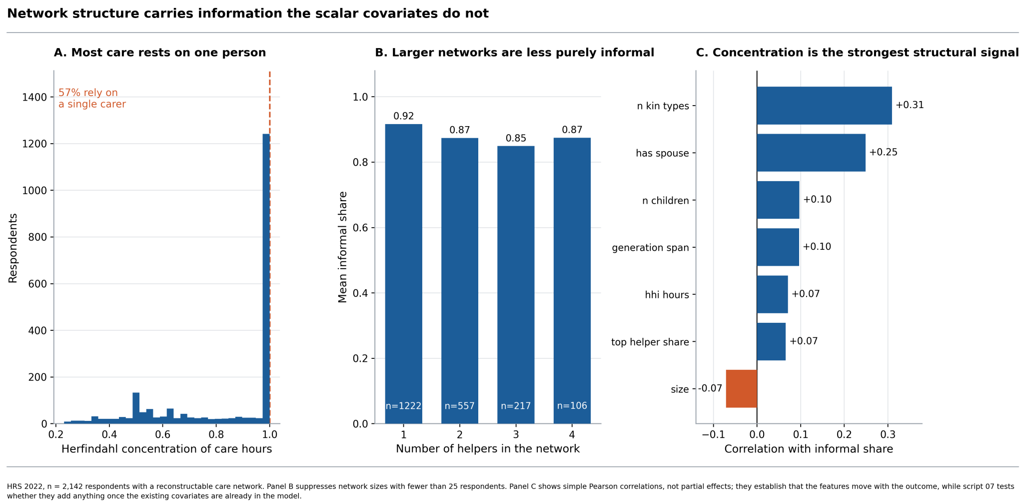


Fig. 12. The features derived from those networks. Panel C reports simple Pearson correlations, not partial effects; they establish that the features move with the outcome, while the ablation in Fig. 14 tests whether they add anything once the existing covariates are already in the model; however, they do not.

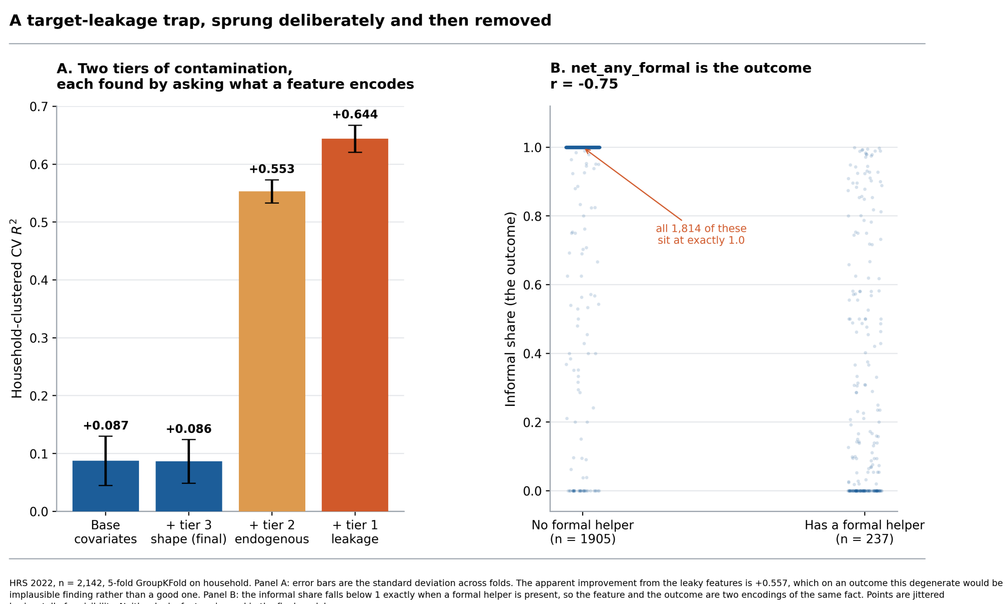
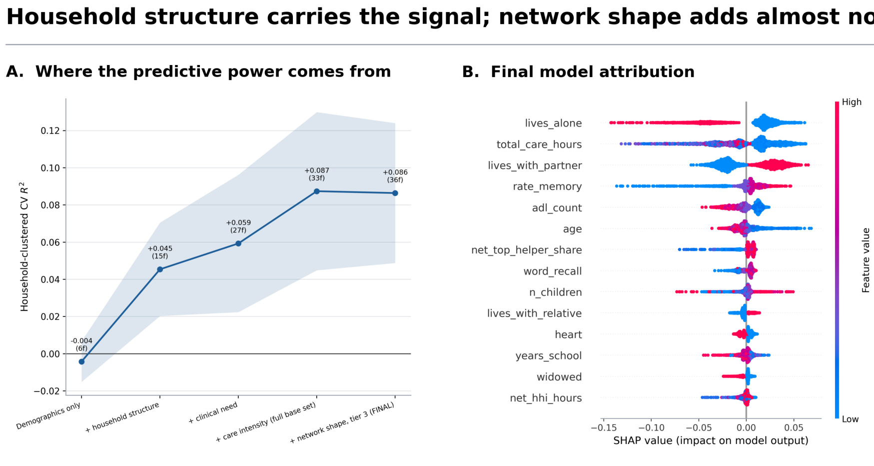


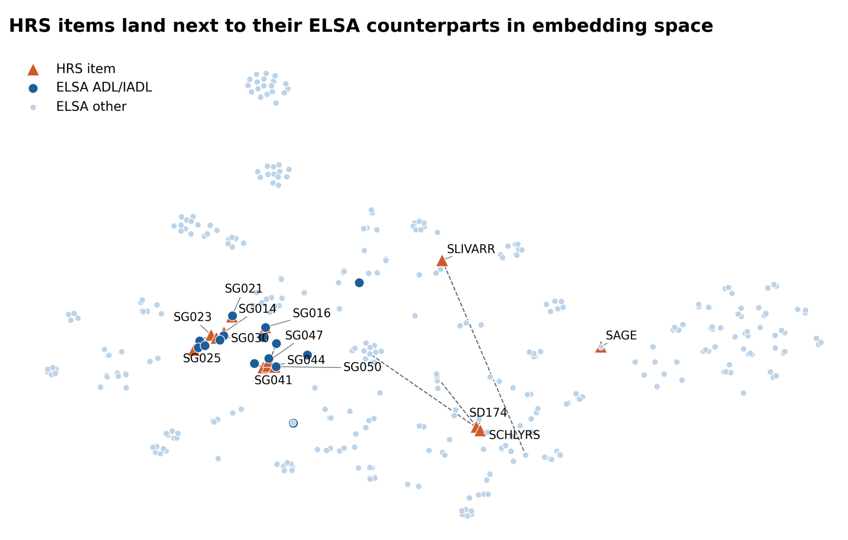
Fig. 13. The two-tier leakage failure of the previous section, in the order it was found. Panel A is the ablation ladder read from the contaminated end downward: adding the tier-1 features lifts household-clustered R^2 to 0.644 and the tier-2 kin-composition features still hold it at 0.553, while the exogenous tier-3 shape features leave the base covariates where they were. An improvement of that size on an outcome this degenerate is more so a consequence than a result. Panel B is the mechanism for tier 1: the informal share drops below one exactly when a formal helper is present, so `net_any_formal` and the outcome are two encodings of one fact. Points are jittered horizontally.



HRS 2022, n = 2,142. Panel A adds feature blocks cumulatively under identical five-fold GroupKFold folds; the shaded band is one standard deviation across folds and is wide enough that the last two steps overlap, so the apparent gain from network structure is not distinguishable from noise. Panel B: each dot is one respondent, horizontal position is that feature's contribution to their prediction, and color is the feature value. Network-structure features receive 6% of total SHAP attribution, so they are real but clearly secondary to living arrangement and care intensity.

Fig. 14. Feature-block ablation with the final model's SHAP summary. Blocks are added cumulatively under identical folds; the shaded band is one standard deviation across folds and is wide enough that the last two steps overlap, so the apparent gain from network structure is not distinguishable from noise.

A.4 The cross-national crosswalk.



t-SNE projection of 60-dimensional LSA embeddings of variable descriptions: 14 HRS items and all 403 ELSA derived variables. Dashed grey lines connect each HRS item to its hand-coded true ELSA counterpart; short lines mean the embedding placed them close together without being told they match. t-SNE distances are not metric, so read adjacency, not scale. The layout is illustrative; the accuracy figures in the following figure are computed on the full-dimensional cosine similarities, not on this projection.

Fig. 15. The embedding space the matcher searches: a t-SNE projection of LSA embeddings of variable descriptions, 14 HRS items against all 403 ELSA derived variables. Dashed lines connect each HRS item to its hand-coded true counterpart; short lines mean the embedding placed them close without being told they match. t-SNE distances are not metric, so read adjacency, not scale.

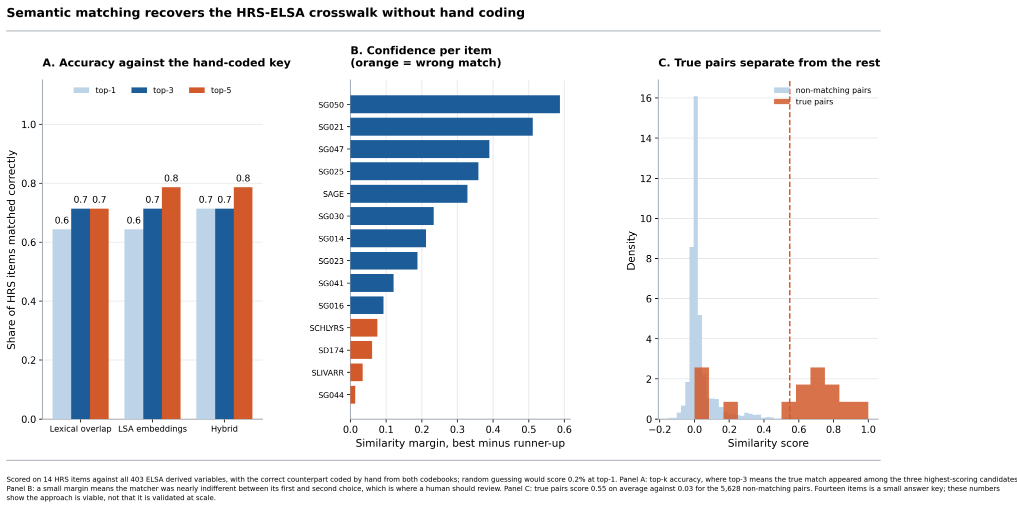


Fig. 16. How well that search does: Panel A scores each matcher against the hand-coded key at top-1, top-3 and top-5, where top-3 means the true counterpart appeared among the three highest-scoring candidates out of 403. Panel B is the per-item similarity margin, the gap between the best and second-best candidate, sorted low to high with incorrect matches in orange; every error falls in the bottom four, which is what makes the margin usable as a rule to sort rather than a diagnostic after the fact. Panel C separates the similarity scores of true pairs from all non-matching pairs. Fourteen items is a small answer key, so this establishes that the approach is viable, not that it is validated at scale.

A.5 Interaction structure.

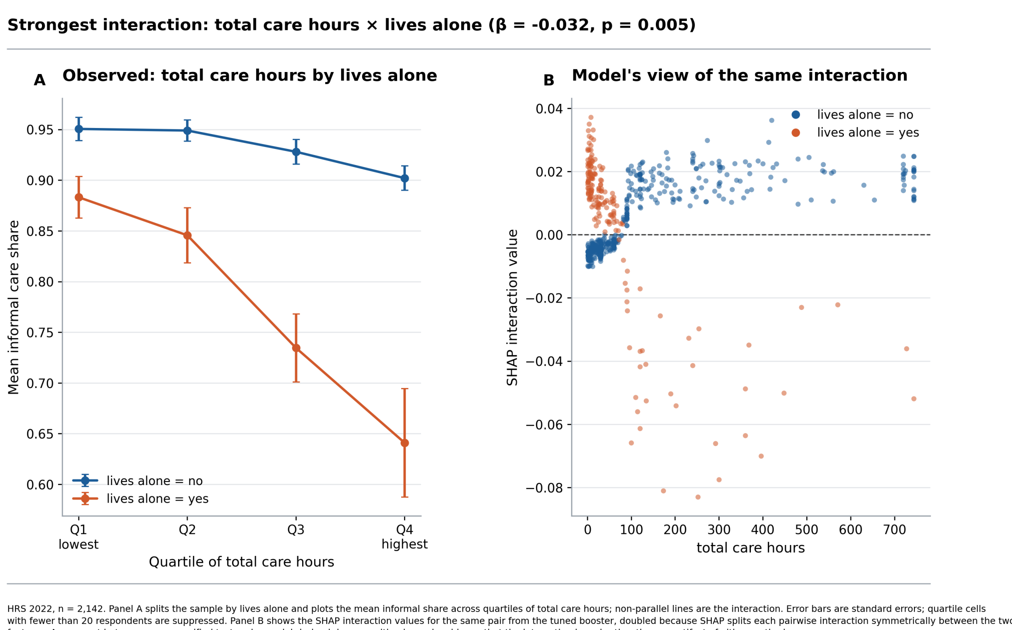


Fig. 17. The strongest interaction decomposed directly. Panel A splits the sample by living arrangement and plots the mean informal share across quartiles of care intensity; non-parallel lines are the interaction. Panel B shows the SHAP interaction values for the same pair from the tuned booster.

A.6 The remaining tables.

The sixteen tables not shown in the body are reproduced here. Where a table runs to every feature, the rows with the largest effect are shown and the full version is the committed CSV. All are regenerated by `run_all.sh`.

Table 4. Descriptives for the continuous analysis variables, HRS 2022, $n = 2{,}142$ care recipients.

Variable	n	Mean	SD	Min	Max	Miss.%
informal_share	2142	0.896	0.282	0	1	0
total_care_hours	2142	125.925	182.615	1	744	0
n_helpers	2142	1.697	1.016	1	8	0
adl_count	2089	2.263	1.913	0	6	2.5
iadl_count	2142	1.739	1.428	0	5	0
age	2142	72.372	12.332	50	101	0
years_school	2129	11.921	3.542	0	17	0.6
word_recall	1697	7.933	3.473	0	20	20.8
rate_memory	1808	3.470	0.988	1	5	15.6
n_children	2142	3.828	2.766	0	22	0

Table 5. Binary analysis variables and the share of the sample at 1.

Variable	Share at 1	n non-missing
female	0.676	2142
lives_alone	0.280	2133
lives_with_partner	0.459	2133
lives_with_relative	0.234	2133
nursing_home	0.034	2142
proxy	0.154	2142
stroke	0.215	2139
dementia	0.129	1986
alzheimers	0.073	2138
neuro_dx	0.345	2142

Table 6. Importance-rank agreement between standardized OLS coefficients and mean absolute SHAP, ordered by the size of the disagreement. Showing 15 of 33 rows; the full table is the committed CSV.

Feature	$\lvert\beta\rvert$	Mean |SHAP|	OLS rk	SHAP rk	Gap
proxy	0.034	0.000	4	28	-24
rate_memory_missing	0.036	0.001	3	25	-22
dementia_missing	0.020	0.000	10	29	-19
total_care_hours	0.007	0.030	20	2	18
alzheimers	0.010	0.000	16	33	-17
years_school	0.004	0.004	28	12	16
n_children	0.006	0.005	24	9	15
iadl_count	0.001	0.002	33	19	14
self_rated_health	0.004	0.002	29	16	13
lives_alone	0.013	0.035	14	1	13
heart	0.006	0.003	27	15	12
nursing_home	0.033	0.003	5	14	-9
diabetes	0.004	0.001	30	21	9
word_recall_missing	0.007	0.000	22	30	-8
kids_within_10mi	0.012	0.001	15	22	-7

Table 7. Bounded against unbounded specifications; only the linear model predicts outside the unit interval.

Specification	Test R^2	Test MAE	Preds. outside $[0, 1]$
Intercept only	-0.007	—	0
OLS	0.011	0.164	58
Fractional logit (GLM binomial)	0.032	0.154	0
Two-part (logit + conditional GLM)	0.035	0.154	0

Table 8. Two-part model coefficients: Part 1 is the probability of any formal care, part 2 the conditional share. Showing 15 of 33 rows; the full table is the committed CSV.

Feature	Part 1 (any formal)	Part 2 (cond. share)	Sign flip
nursing_home	-0.520	-0.854	False
lives_with_partner	-0.481	0.745	True
age	0.421	0.320	False
rate_memory_missing	0.333	0.078	False
lives_with_relative	-0.326	0.369	True
total_care_hours	0.308	0.522	False
adl_count	0.299	-0.126	True
rate_memory	-0.259	0.007	True
race_black	0.249	0.021	False
word_recall_missing	-0.238	-0.441	False
kids_within_10mi_missing	0.212	-0.171	True
kids_within_10mi	-0.188	0.023	True
stroke	0.144	0.014	False
proxy	-0.124	0.187	True
race_other	0.120	0.126	False

Table 9. Naive against household-grouped cross-validation. Grouping scores higher, the opposite of the expected direction, and the gap sits inside fold noise.

Design	Mean R^2	SD across folds	Note
Naive 5-fold KFold (splits individuals)	0.0703	0.0390	leaks 10% of rows across the split
Household GroupKFold (splits households)	0.0874	0.0425	adopted as default; difference from naive is within fold noise

Table 10. Unweighted against survey-weighted coefficients. Weighting drops the 73 nursing-home residents, who carry no weight. Showing 15 of 33 rows; the full table is the committed CSV.

Feature	Unwt. β	Wt. β	Unwt. P	Wt. P	Flips
alzheimers	-0.009	0.007	0.856	0.797	False
widowed	0.001	-0.013	0.944	0.344	False
n_children	-0.001	0.009	0.853	0.220	False
lives_with_relative	0.026	0.037	0.142	0.047	True
hispanic	-0.015	-0.025	0.078	0.004	True
dementia_missing	0.018	0.008	0.728	0.776	False
sep_divorced	-0.001	-0.011	0.917	0.333	False
rate_memory	0.026	0.017	0.000	0.015	False
age	-0.014	-0.005	0.131	0.539	False
lives_alone	-0.023	-0.016	0.197	0.412	False
word_recall	0.012	0.005	0.099	0.524	False
self_rated_health	-0.004	0.003	0.605	0.637	False
race_black	-0.006	0.001	0.461	0.888	False
iadl_count	-0.001	-0.007	0.920	0.384	False
dementia	0.011	0.017	0.157	0.024	True

Table 11. How survey weighting moves the descriptives.

Variable	Unweighted	Weighted	Difference	% change
informal_share	0.891	0.896	0.005	0.5
adl_count	2.178	2.103	-0.075	-3.4
iadl_count	1.723	1.714	-0.009	-0.5
age	74.108	73.362	-0.746	-1.0
lives_alone	0.262	0.223	-0.039	-14.9
lives_with_partner	0.469	0.535	0.066	14.1
nursing_home	0	0	0	—
stroke	0.212	0.205	-0.007	-3.5
dementia	0.131	0.132	0.001	0.6
n_children	3.976	3.757	-0.219	-5.5

Table 12. The nine care-network features. These correlations are what a leakage scan sees, and they do not reveal that six of the nine encode the outcome.

Feature	Mean	SD	Min	Max	r with outcome
net_size	1.697	1.016	1	8	-0.071
net_n_informal	1.566	1.007	0	7	0.213
net_hhi_hours	0.835	0.226	0.230	1	0.071
net_top_helper_share	0.873	0.189	0.250	1	0.066
net_has_spouse	0.392	0.488	0	1	0.249
net_n_children	0.695	0.825	0	6	0.097
net_n_kin_types	1.261	0.608	0	5	0.309
net_generation_span	0.218	0.470	0	3	0.096
net_any_formal	0.111	0.314	0	1	-0.749

Table 13. Neural-network architecture search. No architecture beats the mean, and the best is the smallest.

Architecture	Layers	Units	CV R^2	SD
(8,)	1	8	-0.1703	0.0296
(64, 32, 16)	3	112	-0.1963	0.0438
(64, 32)	2	96	-0.2066	0.0360
(32,)	1	32	-0.2123	0.0568
(128, 64)	2	192	-0.2227	0.0300
(32, 16)	2	48	-0.2656	0.0589
(64,)	1	64	-0.3288	0.0581

Table 14. Model families compared under identical household-grouped folds.

Model family	CV R^2	SD across folds
XGBoost (tuned in script 02)	0.0874	0.0425
Ridge (linear)	0.0614	0.0160
Intercept only	-0.0003	0.0004
MLP (8,), alpha=1	-0.0014	0.0275

Table 15. The ablation ladder, including the two excluded leakage tiers; the last two steps overlap within one fold standard deviation.

Feature block	k	CV R^2	SD
Demographics only	6	-0.0042	0.0109
+ household structure	15	0.0454	0.0251
+ clinical need	27	0.0593	0.0369
+ care intensity (full base set)	33	0.0874	0.0425
+ network shape, tier 3 (FINAL)	36	0.0864	0.0376
+ tier 2 endogenous (excluded)	40	0.5529	0.0199
+ tier 1 leakage (excluded)	35	0.6439	0.0234

Table 16. Final-model SHAP attribution. Showing 15 of 36 rows; the full table is the committed CSV.

Feature	Mean |SHAP|
lives_alone	0.0298
total_care_hours	0.0261
lives_with_partner	0.0260
rate_memory	0.0129
adl_count	0.0112
age	0.0089
net_top_helper_share	0.0067
word_recall	0.0058
n_children	0.0054
lives_with_relative	0.0045
heart	0.0043
years_school	0.0042
widowed	0.0031
net_hhi_hours	0.0030
psych	0.0030

Table 17. The reconstructed HRS–ELSA crosswalk, item by item. All four errors fall in the harder-construct group.

HRS var	ELSA match	Sim.	Correct	Difficulty
SG014	headldr	0.587	True	easy
SG016	headlwa	0.748	True	easy
SG021	headlba	0.812	True	easy
SG023	headlea	0.597	True	easy
SG025	headlbe	0.899	True	easy
SG030	headlwc	0.702	True	easy
SG041	headlpr	0.550	True	easy
SG044	headlme	0.446	False	easy
SG047	headlph	0.826	True	easy
SG050	headlme	1.000	True	easy
SAGE	age	0.708	True	easy
SD174	perid	0.579	False	hard
SCHLYRS	numbuad	0.819	False	hard
SLIVARR	smokerstat	0.453	False	hard

Table 18. Matcher accuracy against the hand-coded key, scored on 14 HRS items against all 403 ELSA derived variables. Random selection scores 0.2% at top-1.

Matcher	Top-1	Top-3	Top-5
Lexical overlap (Jaccard)	0.643	0.714	0.714
LSA embeddings (cosine)	0.643	0.714	0.786
Hybrid (equal weight)	0.714	0.714	0.786

Table 19. The strongest pairwise SHAP interactions, of 630 candidate pairs.

Feature A	Feature B	Mean |interaction|
lives_alone	total_care_hours	0.0065
lives_with_partner	total_care_hours	0.0061
lives_alone	rate_memory	0.0054
age	lives_alone	0.0032
lives_with_partner	net_top_helper_share	0.0027
lives_with_partner	lives_with_relative	0.0024
age	lives_with_partner	0.0022
lives_with_partner	heart	0.0021
years_school	lives_with_partner	0.0021
lives_with_relative	total_care_hours	0.0020
rate_memory	total_care_hours	0.0020
n_children	total_care_hours	0.0020

Agentic Analysis

I chose the research question, the outcome definition, the model families to compare, and the interaction hypotheses; I executed and read every line of code; and I traced every number I report back to a codebook or an output table. Claude was used for a few sparse coding errors throughout, as well as for a couple of questions whose answers were rejected. The space below is for the couple of scripts where the assistant was wrong (the times I had asked for help).

0.0.0.16. Rejected: a harmonization test that scored 100%. The first version of the crosswalk experiment reported perfect top-one accuracy for all three matchers, including a lexical baseline that should not have been competitive; however, inspecting the table explained it. The assistant had written the HRS side of the key itself, as paraphrases of the ELSA wording, and had drawn candidates from only 39 health variables. The matcher was scoring on words the assistant had planted and choosing among a shortlist it had pre-filtered.

How it was wrong: not a coding error. Every line ran, and the reported metric was correctly computed on the data it was given. The failure was in constructing an evaluation that could not produce a negative result. Rewriting the key from verbatim codebook labels and opening the candidate set to all 403 ELSA derived variables dropped accuracy to 71%, which is the number reported here. A confirmatory result had been manufactured by a badly built test, and nothing in the output signaled it.

0.0.0.17. Rejected: a narrative written before the number existed. Asked to examine household clustering, the assistant produced a paragraph explaining that clustering had inflated cross-validated performance and that grouped splitting corrected it; the direction was plausible and the explanation seemed confident. The computation, run afterward by hand, showed grouped cross-validation scoring higher than naive splitting, with the gap smaller than the between-fold standard deviation.

How it was wrong: the explanation preceded the evidence and was fitted to the expected result rather than the observed one. Only 5.3% of households contribute more than one respondent, so there is too little duplication to leak. Grouped splitting is retained as the defensible design, but the paper reports the non-result. Had I accepted the draft, the paper would have contained a fabricated finding.

0.0.0.18. Failure: two-tier target leakage, introduced by the assistant. At first, the nine care-network features were developed with Claude and were not flagged as risky. Including them lifted held-out R^2 from 0.087 to 0.644.

How it was wrong, tier one: `net_any_formal` is true exactly when the informal share is below one. The feature and the outcome are two encodings of the same fact. This was caught in the first review pass, because the name says what it does.

How it was wrong, tier two: the kin-composition features are zero when nobody in the roster is family, which is the same event as the informal share being zero. Ninety-four respondents have `net_n_kin_types = 0` and their mean informal share is 0.000. This survived a review pass and a correlation scan, because no single feature looked unusual on its own. It was found only by looking at each feature and by hand, asking what event makes it zero. The assistant then described the corrected set as “safe”.

The outcome is constructed from the helper roster, so every roster-derived feature is downstream of it by design. The operative test is not “is this correlated with the outcome” but “could this have been known without knowing the outcome.”

0.0.0.19. What this implies. All the errors imply that the output is internally consistent and externally wrong, produced with the same confidence whether right or not. None would have been caught by a test suite, because in each case the code did what it said. Each surfaced only when I asked what a quantity meant rather than whether the script ran. AI assistance moved the bottleneck from writing code to verifying that the code answered the intended question, and the second is the harder job; thus, after these initial asks failed, barring a few errors (as mentioned above), AI assistance was minimized.

0.0.0.20. Website. AI was lastly used to develop the source code for the launched website https://vercel.com/alanna2/care-capacity-site with a descriptive prompt for the overview, most aspects were left blank to insert the graphs/information by hand.

0.0.0.21. Representative transcript. Abridged versions are included in the directory as mentioned above; however, one full representative transcript is also available in the GitHub repository.